

Biotechnology and its Applications

1. The correct sequence of steps followed during gene therapy of an SCID patient is

(i) The retrovirus infects the lymphocyte obtained from the patient and cultured

(ii) cADA gene is inserted into a retrovirus

(iii) The genetically engineered cells are injected into the patient's body

(iv) Retrovirus makes a DNA copy of its RNA along with cADA which is inserted into host DNA

(v) cADA gene is synthesized

(1) (i) → (iii) → (iv) → (v) → (ii)

(2) (v) → (ii) → (i) → (iv) → (iii)

(3) (ii) → (i) → (iv) → (v) → (iii)

(4) (v) → (i) → (ii) → (iv) → (iii)

2. Transgenic animals can be used for/as

(i) Testing of drugs

(ii) Disease models of cancer, Alzheimer, etc.

(iii) Decreasing the resistance to diseases

(iv) Testing vaccine safety

Select the Correct option

(1) (i) and (ii) only

(2) (i), (ii) and (iii)

(3) (ii) and (iv) only

(4) (i), (ii) and (iv)

3. Bacillus thuringiensis forms crystal protein which

(1) Gets activated in acidic medium of insect gut and destroys them

(2) Binds with epithelial cells of midgut after getting activated in alkaline medium

(3) Is functional in the bacterial cell also

(4) Act non-specifically on any insect

4. Root knot disease in a plant is caused by __a__ which can be controlled by __b__.

(1) a-Bacteria ; b-ELISA

(2) a-Nematode ; b-RNA interference

(3) a-Fungus ; b-Blotting

(4) a-Bacteria ; b-PCR

5. In gene therapy for treatment of SCID, in __A__ cells of the patient, the cADA gene is introduced with the help of __B__.

(1) A - Thrombocyte, B - Retrovirus

(2) A - Erythrocyte, B - Arbovirus

(3) A - Lymphocyte, B - Retrovirus

(4) A - Erythrocyte, B - Microinjection

6. ____ method was used to control the infection of a nematode like Meloidogyne incognita in tobacco plant and this control is due to production of ____.

(1) DNAi, ssRNA

(2) RNAi, dsDNA

(3) RNAi, dsRNA

(4) DNAi, dsRNA

7. Which of the following statements is incorrect?

(1) Golden rice is rich in primrose of vitamin A

(2) Human protein (α -1 antitrypsin) obtained from transgenic animals is used to treat emphysema

(3) Human protein enriched milk, which contained the human alpha-lactalbumin was produced by transgenic cow, Molly

(4) A gene cryIAC has been found to be useful in the control of the population of insects like cotton bollworm

8. The Bt-toxin is not toxic to human beings because

- (1) Conversion of pro Bt-toxin to Bt-toxin takes place only in highly alkaline condition
- (2) The pro Bt-toxin activation requires temperature above human body temperature
- (3) The Bt-toxin recognizes only insect specific target
- (4) The Bt-toxin formation from pro Bt-toxin requires pH lower than that present in the human stomach

9. Select the incorrect pair

- (1) Bioremediation – Use of microorganisms to remove pollutants
- (2) Bioethics – Set of standards used to regulate our activities in biological world
- (3) Biopatent – Exclusive right granted for biological entities
- (4) Biopiracy – Use of bioresources by organisations authorised by a national govt

10. Fill in the blanks and choose the correct option.

At present __ (i) __ recombinant therapeutics have been approved globally and __ (ii) __ are presently being marketed in India.

- (1) (i) – 12 ; (ii) – 30
- (2) (i) – 30 ; (ii) – 12
- (3) (i) – 23 ; (ii) – 30
- (4) (i) – 50 ; (ii) – 12

11. RNAi is used to produce transgenic Arabidopsis for resistivity against:

- (1) Lepidopterans
- (2) Viral pathogens
- (3) Nematodes
- (4) Boll worm

12. Match Column - I and Column - II

Column - I

Column - II

- A) Rosie (I) α -1 antitrypsin**
- B) GFP gene (II) Inserted into 1st genetically modified monkey**
- C) Tracy (III) α -lactalbumin**

Options:

- (1) A–II, B–III, C–I
- (2) A–I, B–III, C–II
- (3) A–I, B–II, C–III
- (4) A–II, B–III, C–I

13. Which of the following method can be employed for diagnosis of HIV?

- (A) Schick's test**
- (B) Widal test**
- (C) PCR**
- (D) ELISA**
- (E) Lepromin test**

- (1) A and B
- (2) C and D
- (3) C, D, E
- (4) A, C, D

14. The gene _____ has been introduced in Bt-cotton to protect it from corn borer:

- (1) cry I Ab
- (2) cry I Ac
- (3) cry II Ab
- (4) cry I Ac and cry II Ab

15. Golden rice is:

- (1) Long stored rice having yellow colour tint
- (2) A transgenic rice having gene for β -carotene
- (3) Wild variety of rice with yellow coloured grains
- (4) A transgenic rice enriched with hirudin protein

16. Which of the following is not a transgenic animal?

- (1) Rosie
- (2) Tracy
- (3) Polly
- (4) Dolly

17. Heterologous host for synthesis of humulin (human insulin) is:

- (1) E. coli
- (2) Agrobacterium tumefaciens
- (3) Bacillus thuringiensis
- (4) Arabidopsis

18. SCID is due to deficiency of adenosine deaminase coded by ADA gene. This gene is present on:

- (1) 6th chromosome
- (2) 21st chromosome
- (3) 20th chromosome
- (4) 1st chromosome

19. Which one is incorrect?

- (1) cry-genes of Bacillus thuringiensis encode crystalline insecticidal proteins
- (2) The endotoxin is produced in Bacillus thuringiensis during sporulation
- (3) The inactive protoxin is activated in acidic conditions in foregut of insects
- (4) Hirudin is produced in transgenic plant Brassica napus

20. Which of these is incorrect?

- (1) In humans, insulin is synthesized as a pro-hormone
- (2) Gene for insulin synthesis is on chromosome number 11
- (3) The prohormone i.e. pro-insulin contains an extra stretch called B-peptide
- (4) Insulin extracted from slaughtered cattle and pigs may cause allergy

21. GM plants have been useful in many ways except

- (1) More tolerant to abiotic stresses
- (2) Pest-resistant crops
- (3) Bio-pesticide plants
- (4) Reduced efficiency of mineral usage

22. Choose the odd one w.r.t. Bt toxin

- (1) Insecticidal protein
- (2) Exists as protein crystals
- (3) Exists as inactive protoxins
- (4) Produced continuously during growth period

23. Bt toxin kills certain insects such as tobacco bud worm, army worm etc. These insects belong to which order?

- (1) Coleoptera
- (2) Lepidoptera
- (3) Diptera
- (4) Hymenoptera

24. Read the following statements and choose the correct option.

Statement A: Bt toxin gets activated in alkaline pH of insect gut, which solubilises the protein crystals.

Statement B: Activated toxin binds to the surface of midgut epithelial cells.

- (1) Only statement A is correct
- (2) Only statement B is correct
- (3) Both A and B statements are correct
- (4) Both A and B statements are incorrect

25. The proteins encoded by which gene controls corn borer?

- (1) cry I Ac
- (2) cry II Ab
- (3) cry I Ab
- (4) Sry

26. How many applications given in the box below are application of biotechnology?

Energy production, Waste treatment, Bioremediation, Diagnostics

- (1) Four
- (2) Three
- (3) Two
- (4) One

27. Select the pest resistant plant among the following

- (1) Golden rice
- (2) Flavr Savr tomato
- (3) Transgenic tobacco
- (4) Transgenic Brassica napus

28. Which of the following is incorrect w.r.t. RNAi?

- (1) Present in all eukaryotic organisms
- (2) Method of cellular defense
- (3) Prevent transcription of the gene
- (4) Involves silencing of specific mRNA

29. Nematode-specific genes were introduced into the tobacco plant by using

- (1) Disarmed Ti plasmid of Agrobacterium
- (2) Meloidogyne incognita
- (3) Bacillus thuringiensis
- (4) Retrovirus

30. Eli Lilly, an American company prepared two DNA sequences corresponding to A and B chains of human insulin and introduced them through a plasmid into

- (1) Entamoeba coli
- (2) Escherichia coli
- (3) Salmonella typhimurium
- (4) Haemophilus influenzae

31. Select the treatment method which could be a permanent cure for ADA deficiency in humans

- (1) Enzyme replacement therapy
- (2) Infusion of genetically engineered lymphocytes having functional ADA cDNA

- (3) Bone marrow transplantation after 10 years of age
- (4) Introducing ADA gene into cells at early embryonic stages

32. Which vector was used in the first successful clinical gene therapy in humans?

- (1) Agrobacterium tumefaciens
- (2) Retrovirus
- (3) pBR322
- (4) Bacteriophage

33. Early detection of disease is not possible by performing

- (1) RDT
- (2) Urine analysis
- (3) PCR
- (4) ELISA

34. Which of the following is not correct w.r.t. PCR?

- (1) Routinely used to detect HIV in suspected AIDS patients
- (2) Used to detect mutations in genes
- (3) Used to detect genetic disorders
- (4) Antigen-antibody interaction based test

35. Probe is

- (1) Radioactive dsDNA
- (2) Radioactive dsRNA
- (3) Radioactive ssDNA or ssRNA
- (4) Non-radioactive dsDNA

36. Transgenic models exist for how many human diseases given in the box below?

Cancer, Cystic fibrosis, Rheumatoid arthritis, Alzheimer's

- (1) Four
- (2) Three
- (3) Two
- (4) One

37. Select the incorrect match.

- (1) α -1 antitrypsin — Treatment of emphysema
- (2) α -lactalbumin — Transgenic cow Rosie
- (3) Safety of polio vaccine — Transgenic mice
- (4) Chemical safety testing — Transgenic animals are made less sensitive to toxic substances

38. How many approximate varieties of rice are present in India?

- (1) 27
- (2) 30
- (3) 12
- (4) 2,00,000

39. Select the mismatch.

- (1) Rosie cow — 1997
- (2) Patent on Basmati rice by an American company — 1997
- (3) Human insulin production by Eli Lilly, an American company — 1987
- (4) First clinical gene therapy — 1990

40. Use of bio-resources by multinational companies and other organisations without proper authorization from the countries and people concerned without compensatory payment is known as

- (1) Biopiracy
- (2) Biopatents
- (3) Bioethics
- (4) Bio-resources

41. α -1 antitrypsin is used for the treatment of X whereas Y is used as biological response modifier. X and Y are

- (1) X = Cystic fibrosis, Y = α -interferon
- (2) X = Emphysema, Y = Vincristine
- (3) X = Emphysema, Y = α -interferon
- (4) X = Coronary thrombosis, Y = γ -interferon

42. Choose the correct option

Statement A: About 95% transgenic animals are mice

Statement B: Rosie produced human α -lactalbumin enriched milk.

- (1) Only statement A is correct
- (2) Only statement B is correct
- (3) Both statements A and B are incorrect
- (4) Both statements A and B are correct

43. The Bt toxin does not kill the Bacillus itself because

- (1) Bt toxin is an exotoxin
- (2) It exists as inactive protoxin
- (3) Absence of alkaline medium inside Bacillus
- (4) Absence of acidic medium inside Bacillus

44. Consider the given statement(s)

A. Mature insulin has 51 amino acids

B. In 1983, Eli Lilly prepared human insulin

C. Chain 'A' and Chain 'B' were produced and extracted separately in initial experiments

D. Proinsulin also contains C peptide chain

How many above statement(s) is/are correct?

- (1) One
- (2) Two
- (3) Three
- (4) Four

45. For preparation of humulin, which of the following gene was introduced in early phases?

- (1) Human gene that encodes A, B and C chains of insulin
- (2) Synthetic gene that encodes A and B chains of insulin
- (3) Gene obtained from animals like pig that encode A, B and C chains
- (4) Synthetic gene that encodes A, B and C chains of insulin

46. Genetic engineering has been successfully used for production of

- (1) Transgenic cows to produce milk with only high fat content
- (2) Transgenic mice for safety testing of polio vaccine before its use in humans
- (3) Transgenic sheep to obtain more flesh and milk
- (4) Transgenic animals like bulls for farm work

47. A set of standards that may be used to regulate activities in relation to the biological world while creating GMO are called

- (1) Biopatent
- (2) Biopiracy
- (3) Biowar
- (4) Bioethics

48. Genes required to transform a rice plant into Golden rice enriched in pro vitamin-A were obtained from

- (1) Carrot

- (2) Daffodil
- (3) Sunflower
- (4) Bt cotton and Erwinia

49. Hirudin is obtained from transgenic plant

- (1) *Tobacum nicotiana*
- (2) *Vinca rosea*
- (3) *Brassica napus*
- (4) *Meloidogyne incognita*

50. Select the statement which holds true for cryIAb gene.

- (1) It protects the crops against insects by damaging insect gut lining
- (2) It cannot be inserted into genome of corn plant because corn plant is a monocot
- (3) It codes for those proteins which can kill cotton bollworm
- (4) It cannot kill the insect

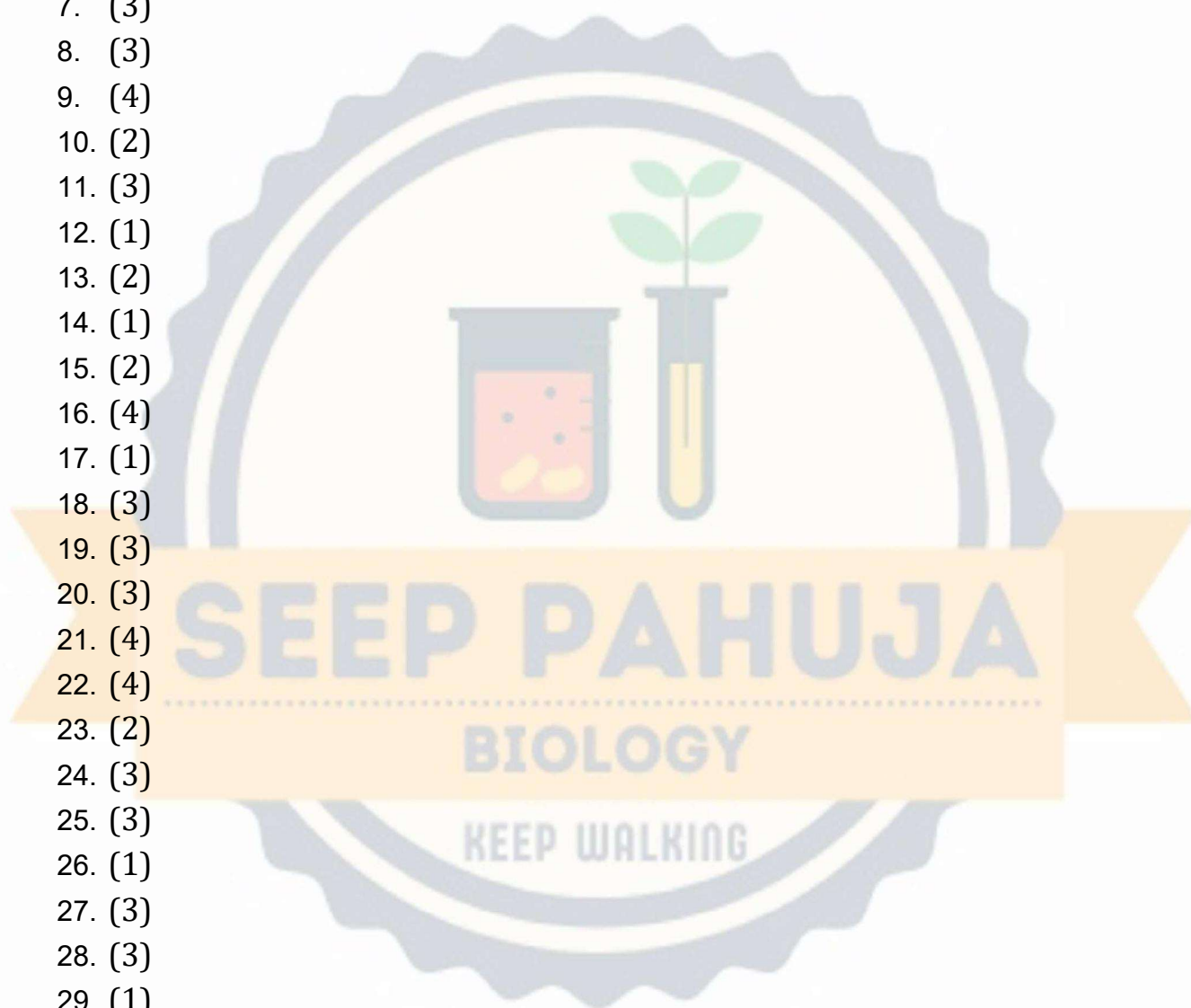
SEEP PAHUJA

BIOLOGY

KEEP WALKING

Answer Key

1. (2)
2. (4)
3. (2)
4. (2)
5. (3)
6. (3)
7. (3)
8. (3)
9. (4)
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- 42. (4)
- 43. (2)
- 44. (4)
- 45. (2)
- 46. (2)
- 47. (4)
- 48. (2)
- 49. (3)
- 50. (1)

